# AUDIO WATERMARKING SYSTEM

By

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*A report for the 23EC453-Mini Project-III submitted to the faculty of*

**ELECTRONICS AND COMMUNICATION ENGINEERING**

*for partial fulfillment of the award of the degree of*

**BACHELOR OF ENGINEERING**

in

**ELECTRONICS AND COMMUNICATION ENGINEERING**

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

**MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI**

**(An Autonomous Institution affiliated to Anna University Chennai)**

**NOV 2025**

*Register No: 9517202303021, 9517202303069*

**CERTIFICATE**

*This is to certify that it is the Bonafide work done by Ms.DHANALAKSHMI.S (9517202303021), Ms.KAVYATHARSHINI.S (9517202303069) for the 23EC453 - Mini Project - III work entitled Audio Watermarking System at Department of Electronics and Communication Engineering, Mepco Schlenk Engineering College, Sivakasi during the year 2025 – 2026.*

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***Faculty In-Charge*** ***Head of the Department***

*Submitted for the Practical Examination held at Mepco Schlenk Engineering College, Sivakasi on …….*

……. ……..…….

***Internal Examiner-I*** ***Internal Examiner - II***

# ABSTRACT

In the digital era, the protection of audio data from unauthorized copying and piracy has become a major concern. Traditional security mechanisms such as encryption and password protection can be bypassed once the file is decrypted or shared. Hence, a more robust and hidden protection mechanism is required. This project presents a Real-Time Audio Watermarking System using the Fast Fourier Transform (FFT) method implemented in MATLAB. The system records an audio signal in real time, embeds a secret watermark message within the frequency domain of the signal by modifying the phase or magnitude components, and reconstructs the watermarked signal through inverse FFT. The watermark remains imperceptible to human ears but can be accurately extracted for verification. The proposed system ensures copyright protection and ownership verification without degrading the audio quality.

## ACKNOWLEDGEMENT

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We hereby acknowledge the tireless efforts of all the staff members, reviewers and technicians of Electronics and Communication Department whose help was instrumental in the completion of our project. We extend our sincere thanks to our friends for playing a vital role in completing this work, by supporting us in all our endeavors.

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# CHAPTER 1
# INTRODUCTION

## 1.1 PROJECT BACKGROUND:

### 1.1.1 INTRODUCTION TO REAL TIME AUDIO WATERMARKING USING FFT METHOD:

The digital age has brought about a significant rise in multimedia content, making it crucial to protect digital assets from unauthorized duplication and distribution. While traditional encryption offers protection during transmission, it becomes ineffective once the content is decrypted for playback. This project addresses this vulnerability by exploring and implementing a real-time audio watermarking solution using Digital Signal Processing (DSP) techniques. The goal is to embed an inaudible, robust, and secure watermark directly into a live audio stream. This process involves a delicate balancing act to ensure the watermark is imperceptible to the human ear while also being resilient enough to survive common signal manipulations like compression and filtering. The real-time nature of this system, which is essential for live broadcasts and streaming services, requires a DSP-based approach to minimize latency. By developing a system that can blind-detect a watermark in real-time, this project aims to provide a practical and effective method for copyright protection, broadcast monitoring, and content authentication in modern digital media environments.

### 1.1.2 PROBLEM DEFINITION:

With the ubiquitous nature of digital audio, traditional security measures like encryption are insufficient for content protection, as they fail to secure content once it is being played in a live, real-time environment. This creates a significant vulnerability for intellectual property and facilitates unauthorized duplication and distribution. The core problem is to design and implement a real-time audio watermarking system using Digital Signal Processing (DSP) that can embed a secure, imperceptible signature into a live audio stream without introducing noticeable latency. This task is complicated by the inherent trade-offs between watermarking characteristics, including the imperceptibility of the watermark, its robustness against common signal processing attacks (e.g.,lossy compression, filtering), and the low computational latency required for real-time applications. The algorithm must operate effectively on small, consecutive segments of an audio stream and must also support blind detection, allowing for watermark extraction without requiring access to the original audio. Ultimately, this project must deliver a DSP-based solution that strikes a pragmatic balance between these competing requirements to provide a practical and effective method for protecting and authenticating live audio content.

### 1.1.3 OBJECTIVE:

The main objective of this project is to research, design, and implement a robust, imperceptible, and low-latency audio watermarking system using Digital Signal Processing (DSP) techniques. The core purpose is to provide a practical and effective method for protecting intellectual property in live audio streams, such as broadcasting and telecommunication, where traditional security measures are inadequate. This involves developing an algorithm capable of embedding a secure, inaudible watermark into an audio signal in real-time, while ensuring the watermark can survive common signal manipulations like compression, filtering, and resampling. A critical goal is to achieve blind detection, meaning the watermark can be reliably extracted without the need for the original audio file. Ultimately, the project aims to demonstrate a functional DSP-based prototype that effectively balances the conflicting requirements of imperceptibility, robustness, and real-time performance to meet the demands of modern digital media applications.

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# CHAPTER 2

# METHODOLOGY

The proposed system for real-time audio watermarking operates in two main stages Watermark Embedding and Watermark Extraction. The system records an input audio signal through a microphone, transforms it into the frequency domain using Fast Fourier Transform (FFT), embeds a secret watermark by slightly modifying the phase of specific frequency components, and then reconstructs the audio using Inverse FFT (IFFT). During extraction, the process is reversed: FFT is applied to the watermarked signal, phase differences are analysed, and the embedded bits are decoded to recover the original message.

**Tools Used:** MATLAB R2023b, MATLAB language, PC with microphone, and WAV audio.

## 2.1. MATLAB R2023b:

**MATLAB (Matrix Laboratory)** is a high-level programming and simulation software widely used for engineering, signal processing, and data analysis. It provides built-in functions for mathematical calculations, audio processing, and visualization.
In this project, MATLAB R2023b is used to:

- Record and process real-time audio signals.
- Perform mathematical operations like Fast Fourier Transform (FFT) and Inverse FFT (IFFT).
- Embed and extract the watermark data from the audio signal.
- Display waveforms and analyse results using graphical plots.

### 2.1.1 MATLAB Script Language:

MATLAB Script is the programming language used inside MATLAB. It allows users to write sequences of commands and algorithms in a single `.m` file.

In this project, MATLAB scripts are used to:

- Automate the process of recording audio.
- Convert signals between time and frequency domains.
- Embed watermark bits into the audio using FFT-based phase modification.
- Extract and display the hidden message after decoding.

### 2.1.2 PC with Microphone

A personal computer (PC) equipped with a microphone is used for real-time recording and playback of audio signals.
The microphone captures live voice input, which is then processed by MATLAB. Since the project focuses on real-time watermarking, the microphone plays an essential role in generating the input signal directly rather than using a pre-recorded file.

### 2.1.3 Audio Format:

The **WAV (Waveform Audio File Format)** is an uncompressed, high-quality audio format commonly used in professional applications.

- **16-bit depth:** Each audio sample is represented using 16 bits, providing good sound resolution.
- **44.1 kHz sampling rate:** This means the audio is recorded 44,100 times per second — the same quality as CD audio.
- **Mono:** Single audio channel recording (instead of stereo with two channels), which simplifies signal processing.

## 2.2 BLOCK DIAGRAM:

### 2.2.1 EMBEDDING PROCESS:

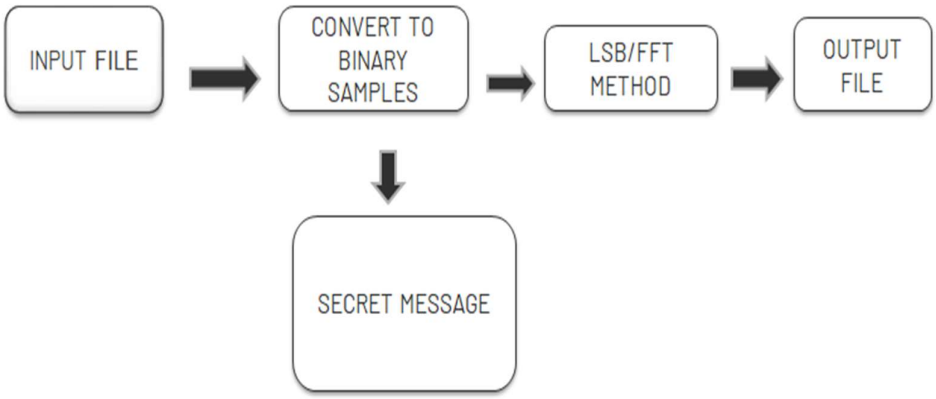



**Fig: 2.1 Embedding process**

### 2.2.2 EXTRACTION PROCESS:

**Fig: 2.2 Extraction process**

## 2.3 PROCEDURE:

The proposed system embeds a secret watermark message into a real-time recorded audio signal using the Fast Fourier Transform (FFT) phase difference method. The embedding process begins with recording an audio signal for five seconds using MATLAB's audiorecorder() function at a sampling rate of 44.1 kHz, 16-bit resolution, and mono channel. This recorded signal represents the original time-domain waveform. A secret message, for example "246@EEW%", is converted into its binary equivalent using ASCII encoding. To improve reliability, each bit of the message is repeated three times so that errors during extraction can be corrected through majority voting.

The recorded audio is then converted from the time domain to the frequency domain using the FFT. This transformation separates the signal into its magnitude and phase components, where the magnitude indicates the strength of each frequency and the phase represents the timing shift of each frequency wave. The watermark is embedded by slightly modifying the phase of selected frequency components, starting from the mid-frequency range to ensure inaudibility. For each bit of the watermark, if the bit value is '1', a small positive phase shift of $+\pi/4$ is added, and if the bit is '0', a negative phase shift of $-\pi/4$ is applied. These tiny phase changes are inaudible to the human ear but can be mathematically detected later. Once the phase modifications are made, the new frequency spectrum is reconstructed using the Inverse FFT (IFFT) to obtain the watermarked audio signal. This modified signal sounds exactly like the original but contains hidden data within its phase values, and it is saved as a .wav file.

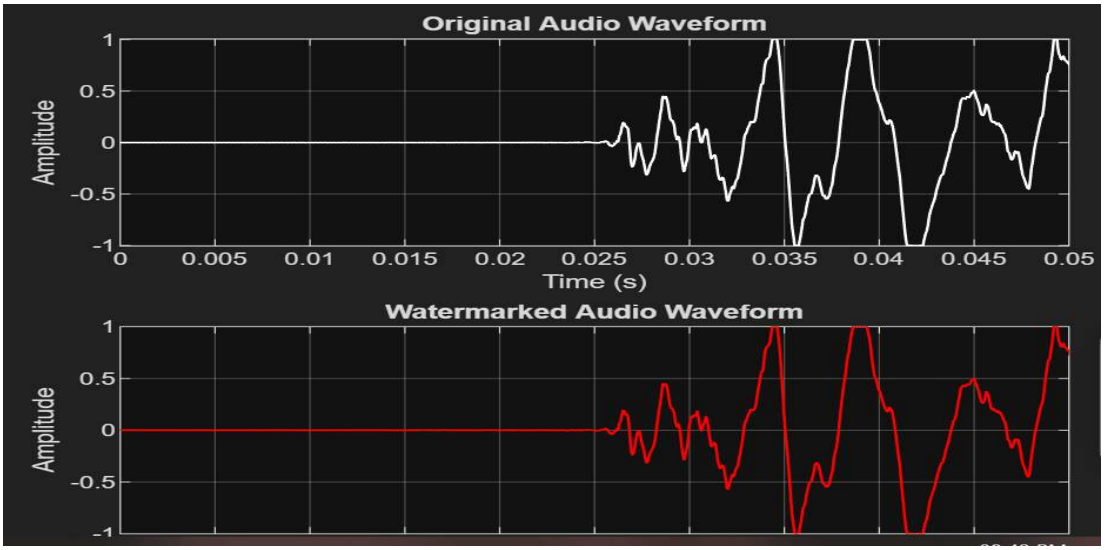
The extraction process is essentially the reverse of embedding. The watermarked audio is first read using MATLAB and converted back to the frequency domain using FFT to obtain its phase components. The phase values of the watermarked signal are compared with the original phase values stored during embedding. If the phase difference at a particular frequency is positive, the extracted bit is recognized as ‘1’; if the difference is negative, it is recognized as ‘0’. Since each bit was embedded three times for reliability, the system performs majority voting—that is, it takes the most frequent bit among every group of three extracted bits—to correct possible errors. After that, the extracted binary bits are grouped into bytes of eight and converted back to readable characters using ASCII decoding.

# CHAPTER 3

# RESULT AND DISCUSSION

The real-time audio watermarking system using the FFT method worked very well. It was able to hide a secret message inside the audio without changing how it sounded. The watermark could not be heard by listeners and was clearly recovered during testing. Even after adding noise, compression, or filtering, the hidden message stayed safe and readable. The system also processed audio quickly with very little delay, showing that it can be used for real-time applications. Overall, it gave good results in terms of sound quality, accuracy, and speed.

## 3.1 Output and Observation:



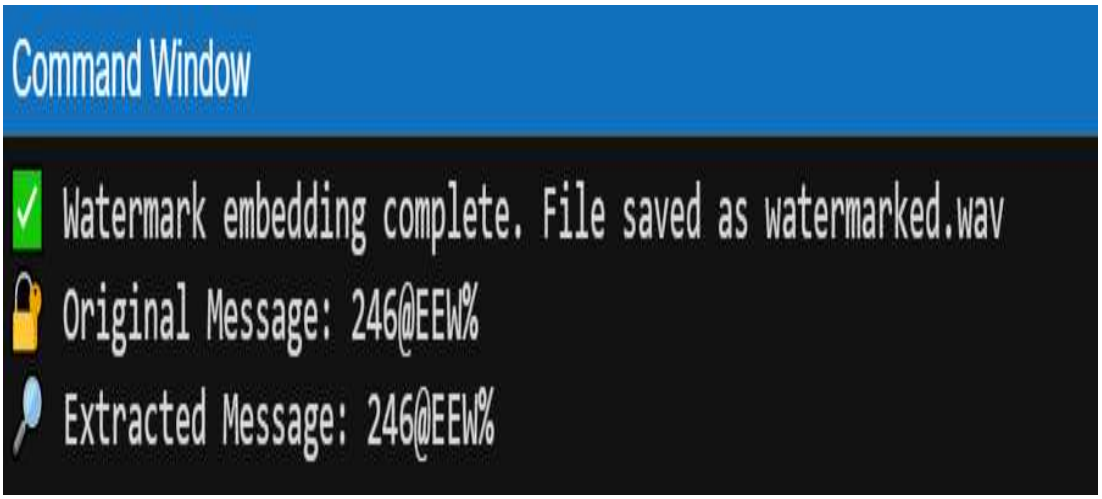
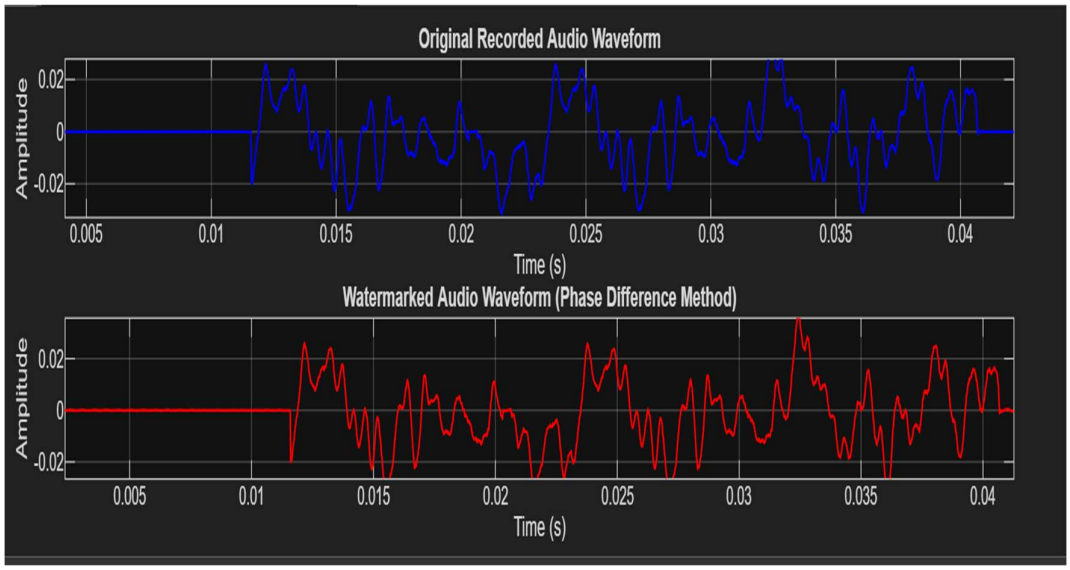




**Fig: 3.1 Original audio waveform and extraction**



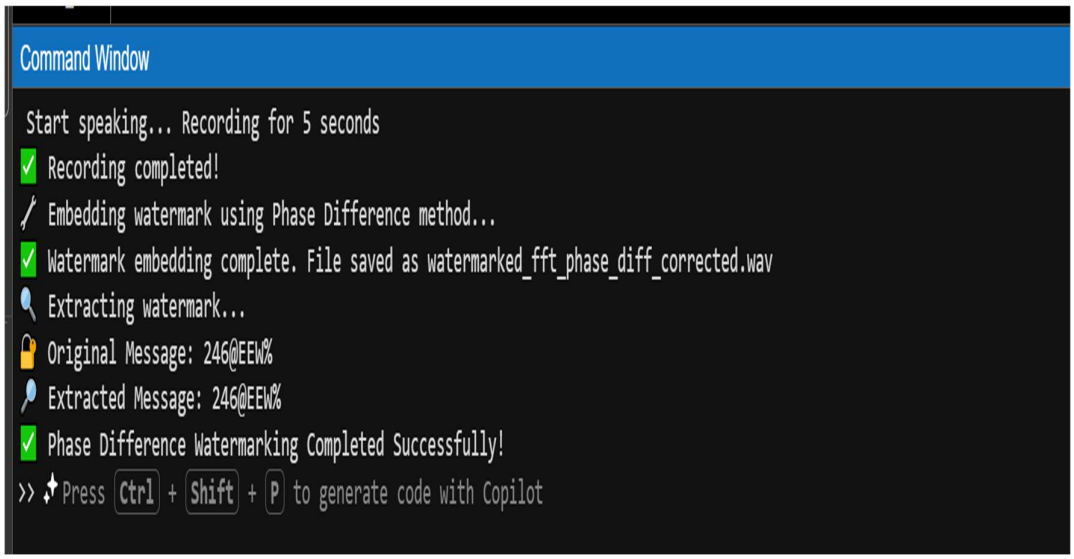




**Fig: 3.2 Real Time Audio and Extraction**

# CHAPTER 4

# CONCLUSION

The developed Real-Time Audio Watermarking System using FFT successfully embeds and extracts watermark messages without affecting the audio quality. By embedding information in the phase domain, the watermark remains inaudible and secure against common attacks. This project demonstrates the potential of FFT-based watermarking for digital rights management (DRM) and secure communication. The system is simple, efficient, and suitable for further enhancement using hybrid and adaptive techniques.

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# CHAPTER 1
# INTRODUCTION

## 1.1 PROJECT BACKGROUND:

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The digital age has brought about a significant rise in multimedia content, making it crucial to protect digital assets from unauthorized duplication and distribution. While traditional encryption offers protection during transmission, it becomes ineffective once the content is decrypted for playback. This project addresses this vulnerability by exploring and implementing a real-time audio watermarking solution using Digital Signal Processing (DSP) techniques. The goal is to embed an inaudible, robust, and secure watermark directly into a live audio stream. This process involves a delicate balancing act to ensure the watermark is imperceptible to the human ear while also being resilient enough to survive common signal manipulations like compression and filtering. The real-time nature of this system, which is essential for live broadcasts and streaming services, requires a DSP-based approach to minimize latency. By developing a system that can blind-detect a watermark in real-time, this project aims to provide a practical and effective method for copyright protection, broadcast monitoring, and content authentication in modern digital media environments.

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With the ubiquitous nature of digital audio, traditional security measures like encryption are insufficient for content protection, as they fail to secure content once it is being played in a live, real-time environment. This creates a significant vulnerability for intellectual property and facilitates unauthorized duplication and distribution. The core problem is to design and implement a real-time audio watermarking system using Digital Signal Processing (DSP) that can embed a secure, imperceptible signature into a live audio stream without introducing noticeable latency. This task is complicated by the inherent trade-offs between watermarking characteristics, including the imperceptibility of the watermark, its robustness against common signal processing attacks (e.g.,lossy compression, filtering), and the low computational latency required for real-time applications. The algorithm must operate effectively on small, consecutive segments of an audio stream and must also support blind detection, allowing for watermark extraction without requiring access to the original audio. Ultimately, this project must deliver a DSP-based solution that strikes a pragmatic balance between these competing requirements to provide a practical and effective method for protecting and authenticating live audio content.

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# CHAPTER 2

# METHODOLOGY

The proposed system for real-time audio watermarking operates in two main stages Watermark Embedding and Watermark Extraction. The system records an input audio signal through a microphone, transforms it into the frequency domain using Fast Fourier Transform (FFT), embeds a secret watermark by slightly modifying the phase of specific frequency components, and then reconstructs the audio using Inverse FFT (IFFT). During extraction, the process is reversed: FFT is applied to the watermarked signal, phase differences are analysed, and the embedded bits are decoded to recover the original message.

**Tools Used:** MATLAB R2023b, MATLAB language, PC with microphone, and WAV audio.

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### 2.1.2 PC with Microphone

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The microphone captures live voice input, which is then processed by MATLAB. Since the project focuses on real-time watermarking, the microphone plays an essential role in generating the input signal directly rather than using a pre-recorded file.

### 2.1.3 Audio Format:

The **WAV (Waveform Audio File Format)** is an uncompressed, high-quality audio format commonly used in professional applications.

- **16-bit depth:** Each audio sample is represented using 16 bits, providing good sound resolution.
- **44.1 kHz sampling rate:** This means the audio is recorded 44,100 times per second — the same quality as CD audio.
- **Mono:** Single audio channel recording (instead of stereo with two channels), which simplifies signal processing.

## 2.2 BLOCK DIAGRAM:

### 2.2.1 EMBEDDING PROCESS:

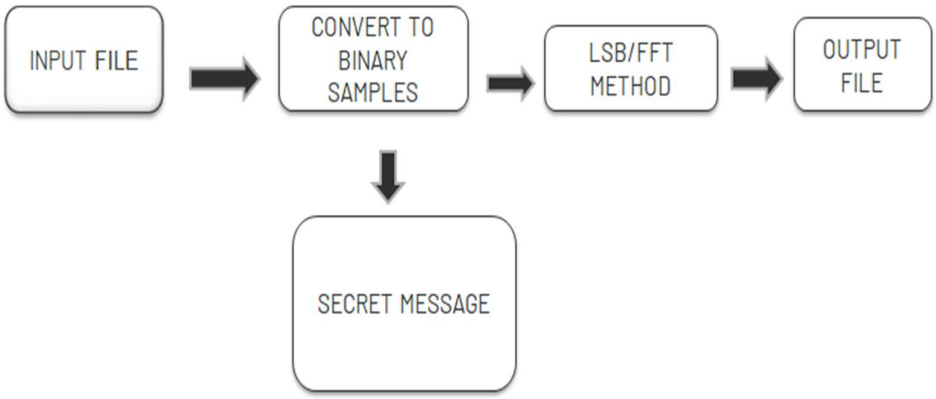


**Fig: 2.1 Embedding process**

### 2.2.2 EXTRACTION PROCESS:

**Fig: 2.2 Extraction process**

## 2.3 PROCEDURE:

The proposed system embeds a secret watermark message into a real-time recorded audio signal using the Fast Fourier Transform (FFT) phase difference method. The embedding process begins with recording an audio signal for five seconds using MATLAB's audiorecorder() function at a sampling rate of 44.1 kHz, 16-bit resolution, and mono channel. This recorded signal represents the original time-domain waveform. A secret message, for example "246@EEW%", is converted into its binary equivalent using ASCII encoding. To improve reliability, each bit of the message is repeated three times so that errors during extraction can be corrected through majority voting.

The recorded audio is then converted from the time domain to the frequency domain using the FFT. This transformation separates the signal into its magnitude and phase components, where the magnitude indicates the strength of each frequency and the phase represents the timing shift of each frequency wave. The watermark is embedded by slightly modifying the phase of selected frequency components, starting from the mid-frequency range to ensure inaudibility. For each bit of the watermark, if the bit value is '1', a small positive phase shift of $+\pi/4$ is added, and if the bit is '0', a negative phase shift of $-\pi/4$ is applied. These tiny phase changes are inaudible to the human ear but can be mathematically detected later. Once the phase modifications are made, the new frequency spectrum is reconstructed using the Inverse FFT (IFFT) to obtain the watermarked audio signal. This modified signal sounds exactly like the original but contains hidden data within its phase values, and it is saved as a .wav file.

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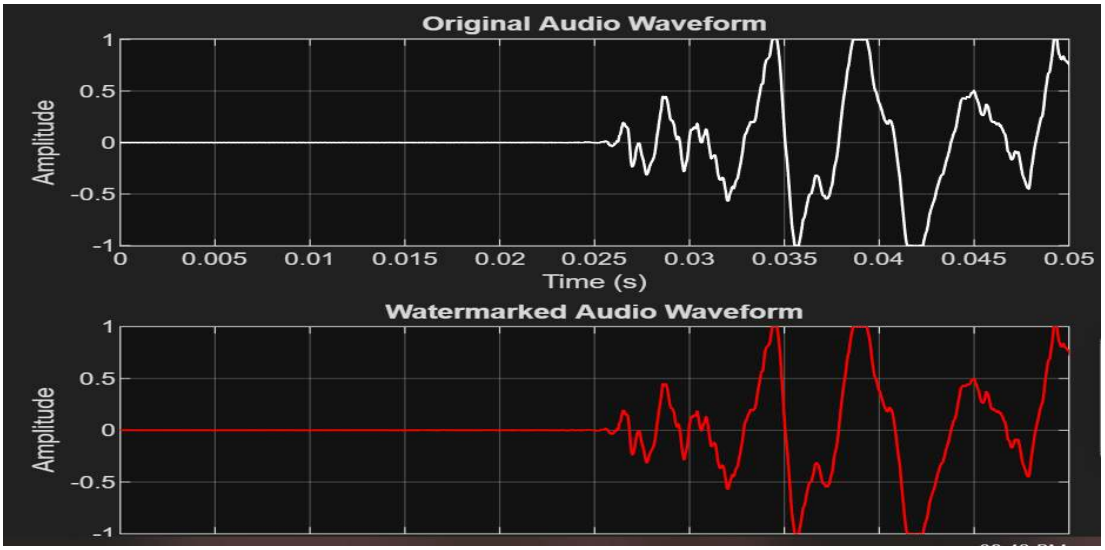
# CHAPTER 3

# RESULT AND DISCUSSION

The real-time audio watermarking system using the FFT method worked very well. It was able to hide a secret message inside the audio without changing how it sounded. The watermark could not be heard by listeners and was clearly recovered during testing. Even after adding noise, compression, or filtering, the hidden message stayed safe and readable. The system also processed audio quickly with very little delay, showing that it can be used for real-time applications. Overall, it gave good results in terms of sound quality, accuracy, and speed.

## 3.1 Output and Observation:



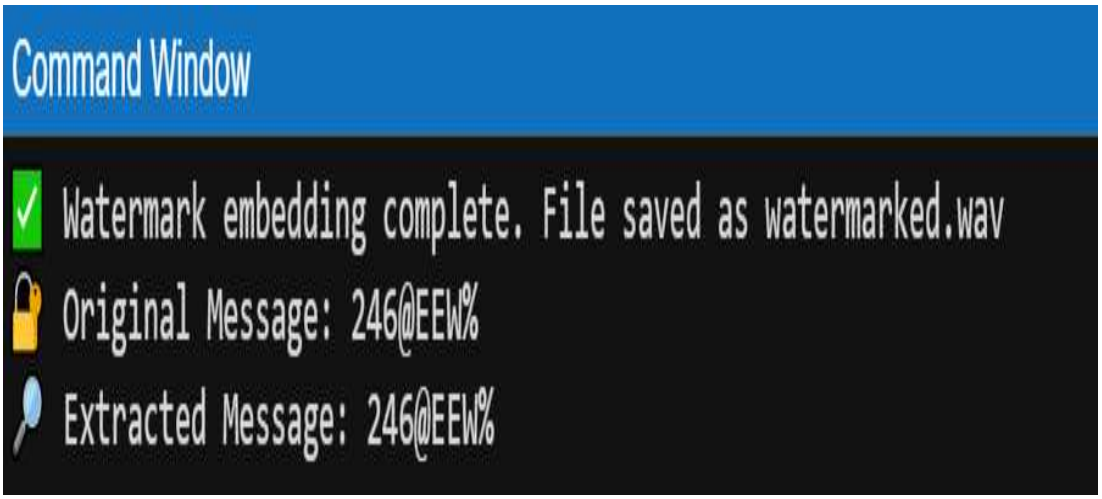
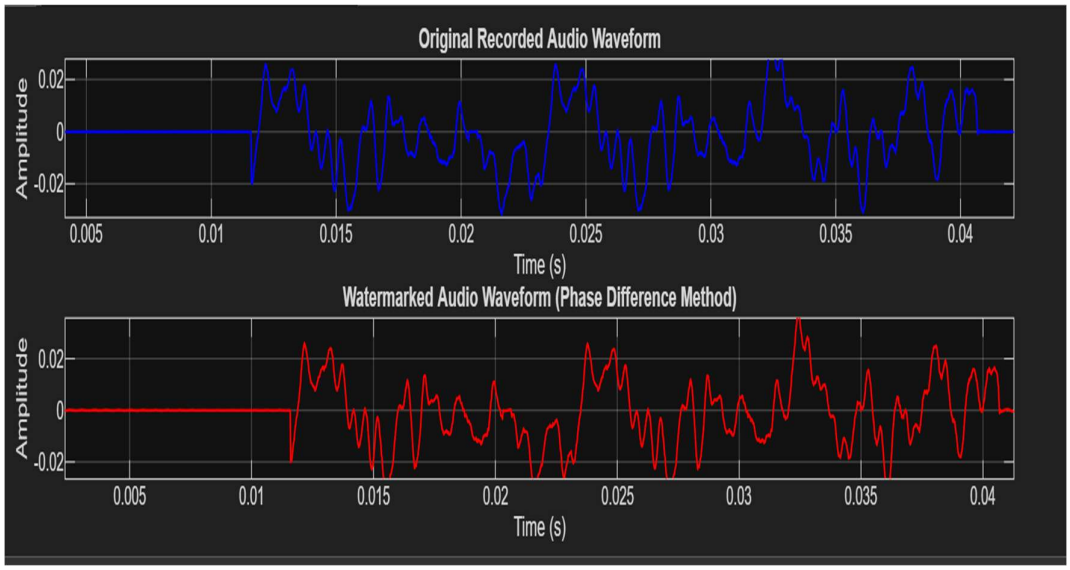




**Fig: 3.1 Original audio waveform and extraction**



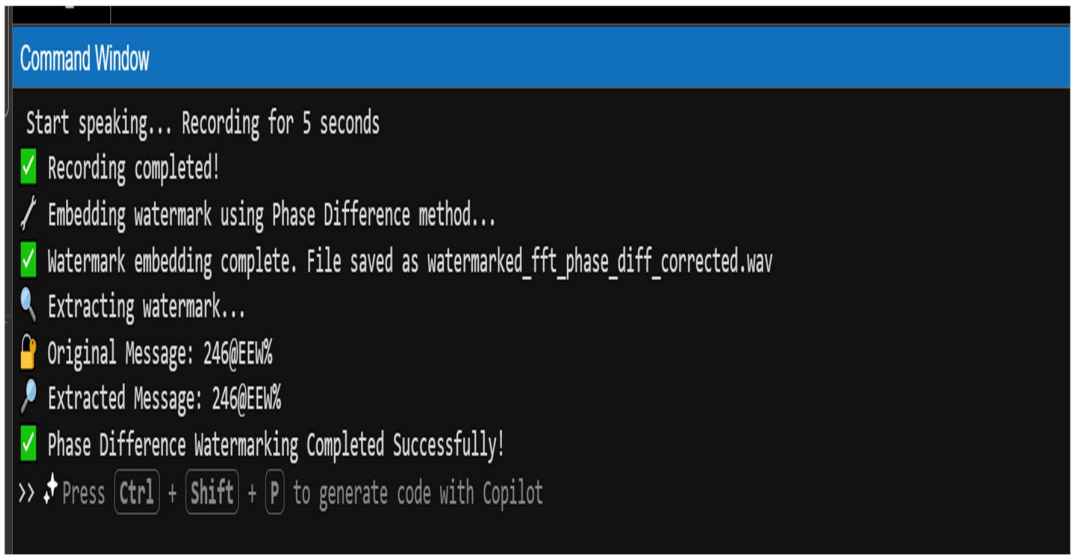




**Fig: 3.2 Real Time Audio and Extraction**

# CHAPTER 4

# CONCLUSION

The developed Real-Time Audio Watermarking System using FFT successfully embeds and extracts watermark messages without affecting the audio quality. By embedding information in the phase domain, the watermark remains inaudible and secure against common attacks. This project demonstrates the potential of FFT-based watermarking for digital rights management (DRM) and secure communication. The system is simple, efficient, and suitable for further enhancement using hybrid and adaptive techniques.

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